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PoseBridge: Bridging the Skeletonization Gap for Zero-Shot Skeleton-Based Action Recognition

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Abstract

Zero-shot skeleton-based action recognition (ZSSAR) is typically treated as a skeleton-text alignment problem: encode joint-coordinate sequences, align them with language, and classify unseen actions. We argue that this alignment is often too late. Skeletons are not complete action observations, but compressed outputs of human pose estimation (HPE); by the time alignment begins, human-object interactions and pose-relative visual cues may no longer be explicit. We call this upstream semantic loss. To address it, we propose **PoseBridge**, an HPE-aware ZSSAR framework that bridges intermediate HPE representations to skeleton-text alignment. Rather than adding an RGB action branch or object detector, PoseBridge extracts pose-anchored semantic cues from the same HPE process that produces skeletons, then transfers them through skeleton-conditioned bridging and semantic prototype adaptation. Across NTU-RGB+D 60/120, PKU-MMD, and Kinetics-200/400, PoseBridge improves ZSSAR performance under the evaluated protocols. On the Kinetics-200/400 PURLS benchmark, which contains in-the-wild videos with diverse scenes and action contexts, PoseBridge shows the clearest separation, improving the strongest compared baseline by 13.3–17.4 points across all eight splits. Our code will be publicly released.

1 Introduction

Skeleton-based action recognition represents actions as compact body-joint trajectories, providing motion-centric cues less tied to appearance variation, background clutter, and illumination changes than raw RGB observations [33, 25, 6]. These properties are particularly attractive for zero-shot skeleton-based action recognition (ZSSAR), where models recognize unseen actions by transferring knowledge through shared motion patterns and textual action descriptions [10, 8, 35, 5, 15]. Most ZSSAR methods [10, 8, 34, 35, 5, 36] follow a skeletonize-then-align (S2A) paradigm: raw visual observations are first reduced to joint coordinates by a human pose estimation (HPE) model, and the resulting skeleton sequences are then aligned with text features in a shared semantic space.

However, this paradigm relies on a rarely examined assumption: skeletons preserve sufficient action semantics for language alignment. In practice, a skeleton is a compressed output of an upstream pose estimator rather than a complete action observation. When raw visual evidence is converted into joint coordinates, cues beyond body motion—such as object presence, human-object interaction, and pose-relative object relations—are no longer explicit. Object-sensitive benchmark actions expose this limitation: as shown in Fig. 1(A), “put on a hat/cap”, “put on headphone”, and “put on glasses” can share similar hand-to-head trajectories, but differ in the manipulated object and its configuration with the actor’s hands and head. As a result, S2A may place similar skeleton features near multiple text prototypes, producing ambiguous zero-shot predictions. We call this mismatch between the semantics required by action text and the evidence retained in skeletons the skeletonization gap, and identify its source as upstream semantic loss before skeleton-text alignment begins.

Preprint.

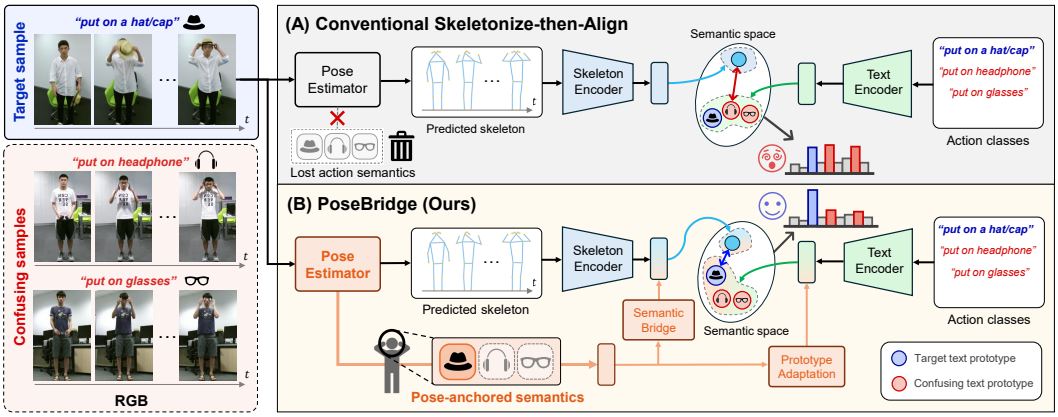


Figure 1: Motivation of PoseBridge. (A) Conventional S2A aligns text with joint-coordinate skeletons after skeletonization, where object, interaction, and pose-relative cues are not explicit. (B) PoseBridge routes HPE-side cues to the alignment space, disambiguating object-sensitive actions with similar skeleton trajectories.

A direct way to mitigate this gap is to augment skeletons with additional context. Language- or prompt-based context can enrich skeleton-text alignment with object and scene descriptions [2, 28], but such context is derived from language or skeleton-side inference rather than recovered from the visual evidence available inside the HPE stream. Another option is to introduce an RGB or object-centric visual pathway [19, 26]. However, this changes the modality protocol of ZSSAR and typically requires an additional visual representation beyond the HPE pass used to obtain skeletons, while also exposing the model to appearance or scene shortcuts. Reusing HPE features is closer to the skeleton extraction process [29], but naive feature fusion treats intermediate maps as generic visual features and does not ensure that the retained cues are action-relevant or organized around the human body. The key challenge is therefore not to append generic context after skeletonization, but to retain and transfer semantics from the skeletonization process in a pose-anchored, skeleton-compatible manner.

To address this challenge, we propose **PoseBridge**, a pose-anchored semantic bridge framework for ZSSAR. As shown in Fig. 1(B), PoseBridge intervenes before skeleton-text alignment by deriving bridge cues from intermediate HPE representations produced during skeleton extraction. Instead of treating HPE as a preprocessing step that only outputs joint coordinates, PoseBridge reinterprets its intermediate representations as pose-anchored semantic cues grounded in the body structure that defines the skeleton. These cues provide object- and pose-relative evidence that joint-coordinate skeletons do not explicitly retain. Importantly, PoseBridge uses cues from the same HPE process that produces skeletons, without an independent RGB action-recognition branch or external object detector.

PoseBridge consists of two coupled stages. First, it learns pose-anchored semantic cues within HPE. Multi-level HPE features are hierarchically refined so that fine-grained visual evidence from shallow layers is injected into deeper pose-structured representations, and body-aware pooling aggregates the refined responses around predicted joints. Auxiliary semantic alignment further encourages the pooled cues to preserve action-relevant evidence rather than generic appearance. Second, PoseBridge transfers these cues to ZSSAR. A skeleton-conditioned semantic bridge lets motion representations query temporal pose-anchored cues, while semantic prototype adaptation calibrates language prototypes toward the pose-semantic space estimated from seen classes. Together, these stages make HPE an upstream source of skeleton-compatible semantics and carry this evidence into skeleton-text alignment.

Experiments evaluate PoseBridge on NTU-RGB+D 60/120, PKU-MMD, and the broader Kinetics-200/400 PURLS benchmark, covering zero-shot learning (ZSL), generalized ZSL harmonic mean (GZSL-H), and random-split protocols. Beyond the commonly used NTU/PKU-style benchmarks, Kinetics tests ZSSAR under larger class diversity and in-the-wild video conditions, where PoseBridge obtains its largest margins.

Our contributions are summarized as follows:

- We identify upstream semantic loss as a limitation of the skeletonize-then-align paradigm, where object, interaction, and pose-relative evidence become non-explicit in joint-coordinate skeletons before skeleton-text alignment begins.
- We propose PoseBridge, a pose-anchored semantic bridge framework that addresses this bottleneck through two coupled stages: semantic preservation within HPE via hierarchical refinement and body-aware pooling, and semantic transfer to ZSSAR via skeleton-conditioned bridging and prototype adaptation.
- PoseBridge achieves the best ZSL accuracy on all standard NTU-RGB+D 60/120 splits and the best GZSL-H among methods without an additional RGB pathway. It further improves random-split GZSL-H by 11.9/12.9 points on NTU-RGB+D 120/PKU-MMD and the strongest baseline under the Kinetics-200/400 PURLS benchmark by 13.3–17.4 points.

2 Related Work

Skeleton-text alignment for ZSSAR. Zero-shot skeleton-based action recognition (ZSSAR) recognizes unseen actions by aligning skeleton representations with textual class semantics [1, 23, 31, 36]. Early methods learn skeleton-text compatibility metrics, e.g., Relation Network [10], while later methods improve the semantic structure of this alignment. SynSE [8] decomposes action labels into verb and noun components, and SMIE [34] uses mutual information estimation for cross-modal matching. Recent methods further refine alignment granularity and robustness: PURLS [35] aligns part-level skeleton features with textual descriptions, SA-DVAE [16] separates action-relevant and irrelevant skeleton components, TDSM [5] denoises skeleton features toward descriptive latent representations, and Flora [3] improves semantic modeling with distribution-aware classification. These methods differ in alignment objectives and semantic representations, but largely operate after skeletonization: joint-coordinate sequences are treated as the primary observation, and semantic alignment is performed only after visual evidence has been compressed into skeletons.

Contextual and pose-estimation cues for skeleton recognition. Joint-coordinate skeletons encode body motion but do not explicitly represent objects, interactions, or scene context. Language-side methods enrich skeleton recognition with external or generated descriptions: Neuron [2] uses LLM-generated contextual side information, and SkeletonContext [28] injects prompt-based object and environment cues. These methods improve class-level semantics, but their context is text-derived and cannot directly recover sample-specific visual evidence made non-explicit during skeleton extraction. Other methods introduce visual information to complement skeletons, such as RGB-skeleton fusion in BSZAR [19] and vision-language alignment in SKI-Models [26]; however, such approaches change the modality protocol by adding an additional visual pathway. A related line uses pose-estimation-side representations rather than joint coordinates alone, including pose estimation maps [21], heatmap volumes [6], and intermediate HPE feature maps in PFMESR [29]. These works show that internal pose-estimation representations can provide richer cues than coordinates, but they are not designed to address the upstream semantic loss in ZSSAR or to align recovered cues with zero-shot text prototypes. PoseBridge instead treats intermediate HPE representations as pose-anchored semantic cues and bridges them to skeleton-text alignment without an independent RGB action-recognition branch or external object detector.

3 PoseBridge

3.1 Problem Setup, Recognition Protocol, and Overview

Given an input video $\mathbf{V} = \{\mathbf{I}_t\}_{t=1}^{T}$, a pose estimator Φ_{HPE} predicts joint coordinates $s_t = \Phi_{\mathrm{HPE}}(\mathbf{I}_t)$, forming the skeleton sequence $\mathbf{S}_i = \{s_t\}_{t=1}^{T}$ for the i-th sample. ZSSAR is trained on seen-class samples $\mathcal{D}^s = \{(\mathbf{S}_i, y_i)\}_{i=1}^{N_s}$ with $y_i \in \mathcal{Y}^s$ and evaluated on disjoint unseen classes $\mathcal{Y}^u$, where $\mathcal{Y}^s \cap \mathcal{Y}^u = \varnothing$. In ZSL, predictions are restricted to $\mathcal{Y}^u$; in GZSL, they are made over $\mathcal{Y}^s \cup \mathcal{Y}^u$.

Recognition protocol. PoseBridge follows an HPE-aware ZSSAR protocol. Raw frames are accessed only through the same HPE process used to obtain skeletons. During ZSSAR training and inference, the HPE model is frozen and provides both skeleton sequence $\mathbf{S}_i$ and cached intermediate

cue sequences $\mathbf{P}_i = \{\mathbf{p}_{i,t}\}_{t=1}^{T}$, where $\mathbf{p}_{i,t}$ denotes a frame-level pose-anchored cue. PoseBridge uses $\mathbf{S}_i$, $\mathbf{P}_i$, and text descriptions; it does not use an independent RGB action-recognition backbone, object detector, optical-flow branch, or unseen-class visual samples for training or prototype construction.

Typical skeletonize-then-align methods encode $\mathbf{S}_i$ as $\mathbf{z}_s = \mathcal{E}_s(\mathbf{S}_i)$ and align it with text prototypes $\mathbf{t}_c = \mathcal{E}_d(\mathbf{d}_c)$, where $\mathcal{E}_s$ and $\mathcal{E}_d$ denote the pretrained skeleton and text encoders. This alignment occurs after visual evidence has been compressed into joint coordinates. PoseBridge instead learns pose-anchored cues from intermediate HPE representations (Sec. 3.2) and transfers them to skeleton representations and prototypes (Sec. 3.3); training and inference are described in Sec. 3.4.

3.2 Learning Pose-Anchored Semantics within HPE

HPE representations contain action-relevant visual evidence that is often lost after skeleton extraction. To preserve it, we convert intermediate HPE features into pose-anchored semantics, as illustrated in Fig. 2. We refine multi-level features, aggregate them around the predicted body structure, and align them with action text semantics while preserving the original pose objective.

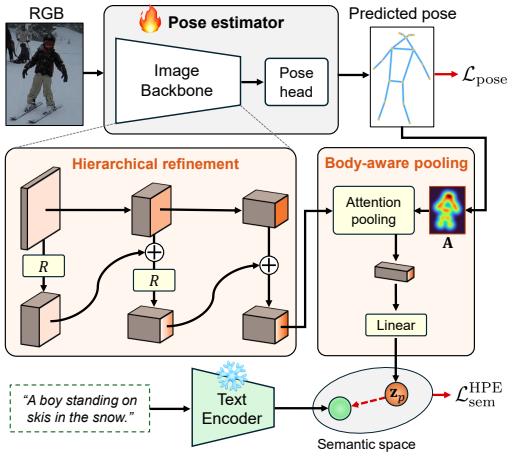


Figure 2: Overview of PoseBridge for learning pose-anchored semantics within HPE.

Hierarchical pose-structured refinement. Intermediate HPE layers provide complementary cues: shallow features retain local appearance and fine-grained spatial details, while deep features encode body configuration and keypoint structure. Using only the deepest feature can therefore discard visual evidence useful for actions requiring fine-grained visual distinctions. We address this by injecting shallower evidence into deeper pose features before semantic pooling.

Let $\{\mathbf{F}^l\}_{l=1}^{L}$ denote shallow-to-deep HPE features of frame $\mathbf{I}$, with $L = 3$ in our experiments. Starting from $\tilde{\mathbf{F}}^1 = \mathbf{F}^1$, each deeper feature is updated as:

$$\tilde{\mathbf{F}}^{l+1} = \mathbf{F}^{l+1} + \alpha\, \mathcal{R}^l\big(\psi^l(\mathcal{P}^l(\tilde{\mathbf{F}}^l))\big), \qquad l = 1, ..., L-1, \tag{1}$$

where $\mathcal{P}^l$ projects the channel dimension and ψ^l resizes the feature to the next-level resolution. $\mathcal{R}^l$ is a convolutional spatial refinement block applied before residual injection, and we fix $\alpha = 0.5$. The final feature $\tilde{\mathbf{F}}^L$ preserves fine-grained visual evidence while maintaining high-level pose structure.

Body-aware semantic pooling. The refined feature map $\tilde{\mathbf{F}}^L$ contains responses over the full spatial map, but useful action semantics should be selected around the actor. We therefore build a body attention map $\mathbf{A}$ from the predicted joint probabilities: joint probabilities are converted into joint heatmaps, averaged over joints, and normalized to form a body-aware spatial prior. The frame-level pose-anchored cue $\mathbf{p}$ is then obtained by body-aware weighted pooling:

$$\mathbf{p} = W_p \left(\frac{\sum_{h,w} \mathbf{A}(h,w)\, \tilde{\mathbf{F}}^L(h,w)}{\sum_{h,w} \mathbf{A}(h,w) + \varepsilon} \right). \tag{2}$$

Here, h, w index spatial locations in $\tilde{\mathbf{F}}^L$, W_p projects the pooled feature into the semantic space, and ε is used for numerical stability. Body-aware pooling encourages $\mathbf{p}$ to emphasize visual evidence near the predicted body structure, including local appearance cues around action-relevant body parts.

Action-semantic alignment. To align pose-anchored semantics with language, we use image-level captions from MS COCO [17] as weak image-language supervision. Each paired description $\mathbf{d}$ is encoded by a pretrained CLIP text encoder $\mathcal{E}_d$ and projected into the same embedding space, yielding $\mathbf{t} = W_d \mathcal{E}_d(\mathbf{d})$. Given a batch of B pose-text pairs, we optimize a CLIP-style symmetric contrastive loss:

$$\mathcal{L}_{\text{sem}}^{\text{HPE}} = -\frac{1}{2B} \sum_{i=1}^{B} \left[\log \frac{\exp(\mathbf{p}_i^\top \mathbf{t}_i / \tau)}{\sum_{j=1}^{B} \exp(\mathbf{p}_i^\top \mathbf{t}_j / \tau)} + \log \frac{\exp(\mathbf{p}_i^\top \mathbf{t}_i / \tau)}{\sum_{j=1}^{B} \exp(\mathbf{p}_j^\top \mathbf{t}_i / \tau)} \right], \tag{3}$$

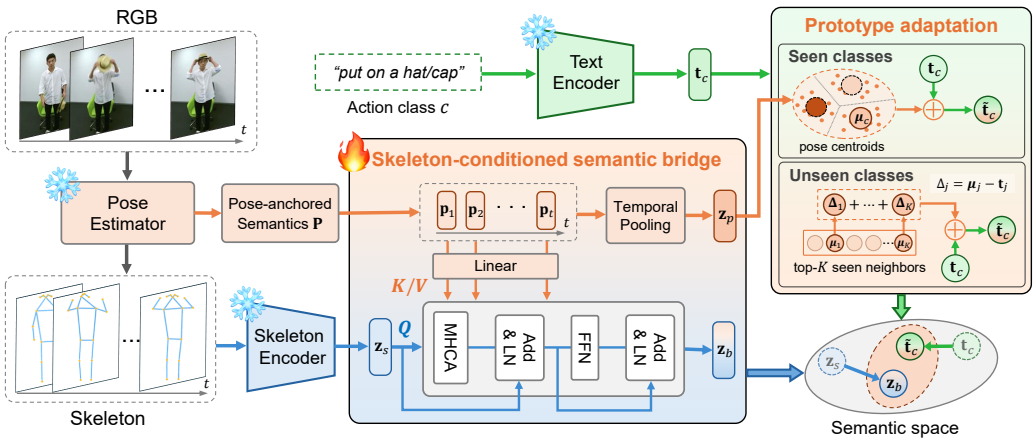


Figure 3: Overview of PoseBridge in the ZSSAR. The pose-anchored semantics extracted from HPE are used for both the skeleton-conditioned semantic bridge and semantic prototype adaptation.

where $\mathbf{p}_i$ and $\mathbf{t}_i$ are L2-normalized embeddings of i-th sample, and $\tau = 0.07$. This contrastive objective aligns pose-anchored cues with textual semantics while separating different samples. The pose estimator is trained with both pose and semantic losses:

$$\mathcal{L}_{\text{HPE}} = \mathcal{L}_{\text{pose}} + \lambda_{\text{hpe}} \mathcal{L}_{\text{sem}}^{\text{HPE}}, \tag{4}$$

where $\mathcal{L}_{\text{pose}}$ is the original pose estimation loss and λ_{hpe} balances the two objectives. After training, the frozen HPE model extracts skeleton sequences $\mathbf{S}_i$ and pose-anchored semantic sequences $\mathbf{P}_i = \{\mathbf{p}_{i,t}\}_{t=1}^{T}$, which are passed to the ZSSAR stage as bridge evidence.

3.3 Bridging Pose-Anchored Semantics to Skeleton-Text Alignment

As illustrated in Fig. 3, we use the pose-anchored semantics learned within HPE to bridge skeleton representations and text prototypes in ZSSAR. The frozen HPE model extracts both the skeleton sequence $\mathbf{S}_i$ and the pose-anchored semantic sequence $\mathbf{P}_i$. PoseBridge uses $\mathbf{P}_i$ for two purposes: enriching the skeleton representation through a skeleton-conditioned semantic bridge and adapting text prototypes with pose-semantic cues. Thus, PoseBridge recovers action-relevant semantics from the skeleton extraction pipeline itself, without introducing an additional RGB recognition branch.

Skeleton-conditioned semantic bridge. Given the skeleton sequence $\mathbf{S}_i$, we first obtain the skeleton representation as $\mathbf{z}_s = W_s(\mathcal{E}_s(\mathbf{S}_i))$, where $\mathcal{E}_s$ is the skeleton encoder and W_s projects the skeleton feature into the semantic embedding space. We then use $\mathbf{z}_s$ as a query and the temporal pose-anchored semantic sequence $\mathbf{P}_i$ as keys and values:

$$\mathbf{z}_b = \text{LN}_2(\mathbf{h}_i + \text{FFN}(\mathbf{h}_i)), \qquad \mathbf{h}_i = \text{LN}_1(\mathbf{z}_s + \sigma(\gamma_b) \odot \text{MHCA}(\mathbf{z}_s, W_b\mathbf{P}_i)), \tag{5}$$

where W_b projects pose-anchored semantics into the bridge embedding space, MHCA denotes multi-head cross-attention, LN is layer normalization, σ is the sigmoid function, and γ_b is a learnable element-wise gate. This injects temporal pose-anchored semantics into the skeleton representation, producing $\mathbf{z}_b$ that remains grounded in skeleton motion while shifting toward a more semantically aligned space. Thus, the bridge restores semantic information lost in coordinate-only skeleton data.

Semantic prototype adaptation. Even after enriching the query with the skeleton-conditioned bridge, the target prototypes remain purely language-defined and may still be misaligned with the pose-semantic space. We therefore adapt text prototypes while keeping them as the primary anchors for zero-shot generality. From the temporal pose-anchored semantic sequence $\mathbf{P}_i$, we obtain a video-level pose-semantic representation $\mathbf{z}_{p,i}$ using attention-based temporal pooling. This converts frame-level HPE semantics into class-level pose evidence for prototype adaptation.

Let $\mathbf{t}_c$ denote the text prototype of class c. For each seen class, we compute the pose-semantic centroid from its training samples and use it for residual prototype adaptation:

$$\tilde{\mathbf{t}}_c = (1-\rho)\mathbf{t}_c + \rho\boldsymbol{\mu}_c, \qquad \boldsymbol{\mu}_c = \frac{1}{|\mathcal{D}_c^s|} \sum_{1 \le i \le |\mathcal{D}_c^s|} \mathbf{z}_{p,i}, \quad c \in \mathcal{Y}^s, \tag{6}$$

where $\mathcal{D}_c^s$ denotes the training samples of seen class c, and ρ is a fixed adaptation coefficient. This preserves the generality of the language prototype while correcting it toward the pose-semantic space.

For unseen classes, no pose samples are available. We therefore transfer the text-to-pose displacement from semantically related seen classes. Let $\mathcal{N}_K(c)$ denote the top-K nearest seen classes to unseen class c in the text prototype space. The unseen prototype is adapted as:

$$\tilde{\mathbf{t}}_c = \mathbf{t}_c + \rho \sum_{j \in \mathcal{N}_K(c)} \omega_{cj}(\boldsymbol{\mu}_j - \mathbf{t}_j), \qquad \omega_{cj} = \frac{\exp(\mathbf{t}_c^\top \mathbf{t}_j / \tau_a)}{\sum_{k \in \mathcal{N}_K(c)} \exp(\mathbf{t}_c^\top \mathbf{t}_k / \tau_a)}, \quad c \in \mathcal{Y}^u. \tag{7}$$

All centroids and adapted prototypes are L2-normalized. Since unseen adaptation uses only residual shifts from nearby seen classes without accessing unseen visual or pose samples, it preserves the zero-shot constraint while making text prototypes more compatible with the pose-semantic space.

3.4 Training Objective and Inference

Training objective. ZSSAR is trained on seen classes only. We optimize three representations with different roles: the skeleton representation $\mathbf{z_s}$, the pose-semantic representation $\mathbf{z_p}$, and the bridge representation $\mathbf{z_b}$. While $\mathbf{z_s}$ and $\mathbf{z_p}$ serve as training anchors for shaping the motion and pose-semantic spaces, $\mathbf{z}_b$ is used as the final representation for zero-shot matching. Let $\mathbf{T}^s = [\mathbf{t}_c]_{c \in \mathcal{Y}^s}$ denote the seen-class text prototype matrix. We define three basic objectives:

$$\mathcal{L}_{\text{cls}}(\mathbf{z}) = \text{CE}(\mathcal{C}(\mathbf{z}), y), \quad \mathcal{L}_{\text{sem}}(\mathbf{z}) = \text{CE}(\mathbf{z}\,(\mathbf{T}^s)^\top, y), \quad \mathcal{L}_{\text{con}}(\mathbf{z}) = \text{SupCon}(\mathbf{z}, y), \tag{8}$$

where $\mathcal{C}$ is a seen-class classifier, CE denotes cross-entropy loss, and SupCon denotes supervised contrastive loss [14], which promotes intra-class compactness and inter-class separation.

For the skeleton and pose anchors, $r \in \{s, p\}$, we use all three losses: $\mathcal{L}_r = \lambda_r^{\text{cls}}\mathcal{L}_{\text{cls}}(\mathbf{z}_r) + \lambda_r^{\text{sem}}\mathcal{L}_{\text{sem}}(\mathbf{z}_r) + \lambda_r^{\text{con}}\mathcal{L}_{\text{con}}(\mathbf{z}_r)$. For the bridge representation, we omit the closed-set classifier and use only semantic matching and contrastive learning: $\mathcal{L}_b = \lambda_b^{\text{sem}}\mathcal{L}_{\text{sem}}(\mathbf{z}_b) + \lambda_b^{\text{con}}\mathcal{L}_{\text{con}}(\mathbf{z}_b)$. This avoids forcing the inference representation $\mathbf{z}_b$ into seen-class classifier boundaries, keeping it better suited for prototype matching with unseen classes. To couple the skeleton and pose-semantic representations, we further use cosine consistency and pose-guided distribution distillation:

$$\mathcal{L}_{\text{align}} = \lambda_{\text{s2p}}\,(1 - \cos(\mathbf{z}_s, \mathbf{z}_p)) + \lambda_{\text{kd}}\,\text{KL}\left(\text{softmax}\Big(\frac{\mathbf{z}_p(\mathbf{T}^s)^\top}{\tau_d}\Big)^{\text{sg}} \middle\| \text{softmax}\Big(\frac{\mathbf{z}_s(\mathbf{T}^s)^\top}{\tau_d}\Big)\right), \tag{9}$$

where KL denotes Kullback–Leibler divergence loss, τ_d is the distillation temperature, and sg denotes stop-gradient on the pose-guided teacher distribution. The final objective is $\mathcal{L}_{\text{ZSSAR}} = \mathcal{L}_s + \mathcal{L}_p + \mathcal{L}_b + \mathcal{L}_{\text{align}}$. Overall, this objective shapes $\mathbf{z}_s$ and $\mathbf{z}_p$ into discriminative semantic anchors, while training $\mathbf{z}_b$ as an open prototype-matching representation for unseen-class transfer.

Inference. At inference, we use the bridge representation $\mathbf{z}_b$ as the final query and match it with the adapted prototypes $\tilde{\mathbf{t}}_c$. Since both are L2-normalized, similarity is computed by dot product, and ZSL/GZSL predictions are computed as:

$$\hat{y}_{\text{zsl}} = \arg\max_{c \in \mathcal{Y}^u} \mathbf{z}_b^\top \tilde{\mathbf{t}}_c, \qquad \hat{y}_{\text{gzsl}} = \arg\max_{c \in \mathcal{Y}^s \cup \mathcal{Y}^u} \big(\mathbf{z}_b^\top \tilde{\mathbf{t}}_c - \kappa\,\mathbf{1}[c \in \mathcal{Y}^s]\big), \tag{10}$$

where κ is a seen-class calibration coefficient that mitigates seen-class bias in GZSL, following prior work [2]. Thus, inference explicitly couples the two sides of our bridge: $\mathbf{z}_b$ carries pose-anchored semantics recovered from the skeleton extraction process, while $\tilde{\mathbf{t}}$ provides skeleton-compatible semantic targets, jointly reducing the skeleton-text semantic gap.

4 Experiments

4.1 Experimental Setup

Datasets. We evaluate PoseBridge on NTU-RGB+D 60 [24], NTU-RGB+D 120 [20], PKU-MMD [18], and Kinetics-200/400 [12, 35]. NTU-RGB+D 60/120 and PKU-MMD are controlled RGB-D skeleton action benchmarks, while Kinetics-200/400 provides in-the-wild videos with diverse scenes and action contexts. For all datasets, we follow the standard seen/unseen class splits used in prior ZSSAR works [8, 34, 35]. Detailed dataset descriptions are provided in Appendix B.

Table 1: Comparison on NTU-RGB+D 60/120 under standard ZSL/GZSL splits. † denotes methods using an additional RGB pathway with a pretrained CLIP visual encoder. The best results are highlighted in **red**, and the second-best results are blue.

Method	Venue	NTU-RGB+D 60 (Xsub)								NTU-RGB+D 120 (Xsub)							
		55/5 Split				48/12 Split				110/10 Split				96/24 Split			
		ZSL	GZSL			ZSL	GZSL			ZSL	GZSL			ZSL	GZSL		
		Acc	*S*	*U*	*H*	*Acc*	*S*	*U*	*H*	*Acc*	*S*	*U*	*H*	*Acc*	*S*	*U*	*H*
ReViSE [9]	ICCV 2017	53.9	74.2	34.7	29.2	17.5	62.4	20.8	31.2	55.0	48.7	44.8	46.7	32.4	49.7	25.1	33.3
JPoSE [30]	ICCV 2019	64.8	64.4	50.3	56.5	28.8	60.5	20.6	30.8	51.9	47.7	46.4	47.0	32.4	38.6	22.8	28.7
CADA-VAE [23]	CVPR 2019	76.8	69.4	61.8	65.4	29.0	51.3	27.0	35.4	59.5	47.2	19.8	48.4	35.8	41.1	34.1	37.3
SynSE [8]	ICIP 2021	75.8	61.3	56.9	59.0	33.3	52.2	27.9	36.3	62.7	52.5	57.6	54.9	38.7	56.4	32.2	41.0
SMIE [34]	ACMMM 2023	78.0	-	-	-	40.2	-	-	-	65.7	-	-	-	45.3	-	-	-
PURLS [35]	CVPR 2024	79.2	-	-	-	41.0	-	-	-	72.0	-	-	-	52.0	-	-	-
SA-DVAE [16]	ECCV 2024	82.4	62.3	70.8	66.3	41.4	50.2	36.9	42.6	68.8	61.1	59.8	60.4	46.1	58.8	35.8	44.5
STAR [1]	ACMMM 2024	81.4	69.0	69.9	69.4	45.1	62.7	37.0	46.6	63.3	59.9	52.7	56.1	44.3	51.2	36.9	42.9
SCoPLe [36]	CVPR 2025	84.1	69.6	71.9	70.8	53.0	54.5	61.8	57.9	74.5	63.5	61.1	62.3	52.2	53.3	51.2	52.2
Neuron [2]	CVPR 2025	86.9	69.1	73.8	71.4	62.7	61.6	56.8	59.1	71.5	67.6	59.5	63.3	57.1	67.5	44.4	53.6
TDSM [5]	ICCV 2025	86.5	-	-	-	56.0	-	-	-	74.2	-	-	-	65.1	-	-	-
FS-VAE [31]	ICCV 2025	86.9	77.0	74.5	75.7	57.2	56.2	48.6	52.1	74.4	59.2	67.9	63.3	62.5	57.8	51.9	54.7
BSZSL† [19]	ESWA 2025	83.0	85.1	72.0	78.0	53.0	73.9	36.5	48.8	77.7	80.8	71.4	**75.8**	56.1	72.1	53.8	61.6
SKI-VLM† [26]	AAAI 2025	82.2	-	-	-	52.0	-	-	-	77.5	-	-	-	59.3	-	-	-
PoseBridge	**This work**	**88.8**	82.0	84.0	**83.0**	**73.2**	75.6	61.9	**68.0**	**80.3**	71.3	65.0	68.0	**70.9**	66.1	59.4	**62.6**

Table 2: Comparison on NTU-RGB+D 60/120 and PKU-MMD under random ZSL/GZSL splits.

Method	Venue	NTU-RGB+D 60 (Xsub)		NTU-RGB+D 120 (Xsub)		PKU-MMD (Xsub)	
		55/5 Split		110/10 Split		46/5 Split	
		ZSL	GZSL (*H*)	ZSL	GZSL (*H*)	ZSL	GZSL (*H*)
ReViSE [9]	ICCV 2017	60.9	60.3	44.9	40.3	59.3	49.8
JPoSE [30]	ICCV 2019	59.4	60.1	46.7	43.7	57.2	51.6
CADA-VAE [23]	CVPR 2019	61.8	66.4	45.2	45.6	60.7	45.8
SynSE [8]	ICIP 2021	64.2	67.5	47.3	43.5	53.9	49.5
SMIE [34]	ACMMM 2023	65.1	-	46.4	-	60.8	-
SA-DVAE [16]	ECCV 2024	84.2	75.3	50.7	47.5	66.5	54.7
SCoPLe [36]	CVPR 2025	83.7	77.7	53.3	54.1	71.4	54.9
TDSM [5]	ICCV 2025	88.9	-	69.5	-	70.8	-
FS-VAE [31]	ICCV 2025	-	-	-	-	71.2	59.0
PoseBridge	**This work**	**90.5**	**78.2**	**74.8**	**66.0**	**77.5**	**71.9**

Evaluation protocol. For zero-shot learning (ZSL), we report top-1 accuracy on unseen classes. For generalized zero-shot learning (GZSL), the prediction space contains both seen and unseen classes. Following prior works, we report the seen-class accuracy S, the unseen-class accuracy U, and their harmonic mean H, which measures the balance between seen and unseen recognition.

Implementation details. We use RTMPose [11] as the 2D pose estimator. The HPE model is trained with the original pose estimation objective and our action-semantic alignment objective on the MS COCO dataset [17], and is then frozen to extract both estimated 2D skeleton sequences and pose-anchored semantics used in the ZSSAR stage. Following common ZSSAR evaluation protocols for fair comparison, we adopt Shift-GCN [4] as the skeleton encoder and use the pretrained CLIP text encoder [22] to obtain semantic prototypes from action descriptions. All models are trained only on seen classes and evaluated on disjoint unseen classes. Detailed implementation settings are provided in Appendix A, and hyperparameter/design choice analyses are presented in Appendix D.

4.2 Comparison with State-of-the-Art

Evaluation on standard split benchmark. We compare PoseBridge with state-of-the-art ZSSAR methods on the standard NTU-RGB+D 60/120 Xsub protocols, following [8]. As shown in Table 1, PoseBridge achieves the best ZSL performance across all settings and substantially improves GZSL performance. The gains are especially clear under harder splits with fewer seen and more unseen classes: on NTU-RGB+D 60 under the 48/12 split, PoseBridge improves the previous best ZSL

Table 3: Results on Kinetics-200/400 under the PURLS [35] ZSL benchmark. * indicates the use of multiple text prompts; otherwise, a single text prompt is used.

Method	Venue	Kinetics-200				Kinetics-400			
		180/20 Split	160/40 Split	140/60 Split	120/80 Split	360/40 Split	320/80 Split	300/100 Split	280/120 Split
DeVISE [7]	NeurIPS 2013	22.2	12.3	8.0	5.7	18.4	10.2	9.5	8.3
ReViSE [9]	ICCV 2017	25.0	13.3	8.1	6.2	20.8	11.8	9.5	8.2
PURLS [35]	CVPR 2024	26.0	15.9	10.2	7.8	22.5	15.1	11.4	11.0
PURLS* [35]	CVPR 2024	32.2	22.6	12.0	11.8	34.5	24.3	17.0	14.3
TDSM [5]	ICCV 2025	38.2	24.4	15.3	13.1	38.9	26.2	18.5	16.1
PoseBridge	**This work**	**55.6**	**38.3**	**28.6**	**26.4**	**52.3**	**39.6**	**32.9**	**32.1**

Table 4: Accuracy-efficiency comparison on NTU-RGB+D 60/120 at the ZSSAR recognition stage. We report ZSL accuracy under standard splits, with trainable/total parameters, GFLOPs, and FPS.

Method	Venue	NTU-RGB+D 60 (Xsub)		NTU-RGB+D 120 (Xsub)		Trainable / Total #Params (M)	GFLOPs (G)	FPS
		55/5 Split	48/12 Split	110/10 Split	96/24 Split			
Neuron [2]	CVPR 2025	86.9	62.7	71.5	57.1	2.3 / 126.7	7.5	1165.4
TDSM [5]	ICCV 2025	86.5	56.0	74.2	65.1	260.4 / 384.2	10.6	772.5
FS-VAE [31]	ICCV 2025	86.9	57.2	74.4	62.5	**0.2 / 64.4**	**3.7**	1237.0
SKI-VLM [26]	AAAI 2025	82.2	52.0	77.5	59.3	2.8 / 215.9	281.0	22.6
PoseBridge	**This work**	**88.8**	**73.2**	**80.3**	**70.9**	3.8 / 128.2	7.3	**1398.0**

accuracy by 10.5 points and the GZSL harmonic mean by 8.9 points. The strong results on NTU-RGB+D 120 under the 96/24 split further confirm the effectiveness of pose-anchored semantics for semantically diverse unseen actions.

Notably, PoseBridge outperforms recent RGB-assisted methods in almost all settings, despite using no additional RGB recognition branch. Instead, it recovers action-relevant visual semantics from intermediate HPE representations already computed during skeleton extraction, showing that the gain comes from better exploiting the skeletonization process rather than adding heavy RGB-based modules. To address potential concerns about skeleton input formats, Appendix C provides a controlled comparison using the same RTMPose-extracted 2D skeletons, where PoseBridge remains superior, along with additional Xview and Xset results.

Evaluation on random split benchmark. Table 2 reports results under the random split protocol introduced by SMIE [34], where three randomly selected seen/unseen splits are evaluated and averaged. PoseBridge again delivers consistent improvements across NTU-RGB+D 60, NTU-RGB+D 120, and PKU-MMD datasets. These results demonstrate that the effectiveness of PoseBridge is consistent across different seen-unseen partitions, indicating that the proposed semantic bridge improves ZSSAR beyond a specific split configuration.

Evaluation on Kinetics-200/400 dataset. We further evaluate PoseBridge on the more challenging Kinetics-200/400 benchmarks following PURLS [35]. Unlike NTU-RGB+D and PKU-MMD, Kinetics contains in-the-wild videos with diverse scenes, viewpoints, subjects, and action contexts. As shown in Table 3, PoseBridge substantially outperforms prior methods across all splits. In particular, PoseBridge improves Kinetics-200 under the 180/20 split by 17.4 points, from 38.2% to 55.6%, and Kinetics-400 under the 280/120 split by 16.0 points, from 16.1% to 32.1%. These large gains on in-the-wild videos strongly support our motivation that preserving action-relevant semantics during skeletonization is important for ZSSAR.

Computational cost. PoseBridge achieves a strong accuracy-efficiency trade-off among recent methods. As shown in Table 4, it obtains the best accuracy on all NTU-RGB+D 60/120 splits while achieving the fastest inference speed under the same ZSSAR recognition-stage evaluation setting. For all methods, trainable/total parameters, GFLOPs, and FPS are measured with the skeleton encoder and text encoder included on a single NVIDIA RTX A6000 GPU. Compared with SKI-VLM [26], which relies on an additional RGB visual encoder and incurs substantially larger GFLOPs, PoseBridge recovers action-relevant semantics from intermediate HPE representations with a much more compact recognition pipeline. TDSM has the largest parameter count due to its diffusion-based backbone and text-conditioning components. Overall, PoseBridge improves zero-shot recognition while preserving the efficiency of a skeleton-based pipeline. Component-wise costs for the full PoseBridge pipeline, from pose estimation to ZSSAR prediction, are provided in Appendix D.3.

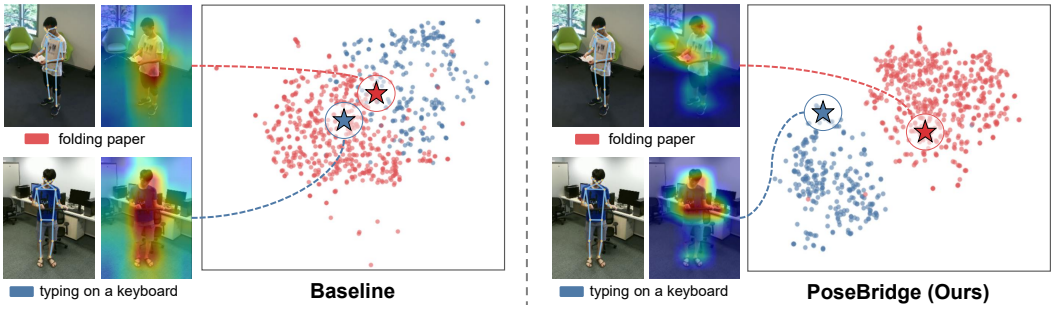


Figure 4: Qualitative comparison between the baseline and PoseBridge on the confusing action pair "folding paper" and "typing on a keyboard". We visualize predicted skeletons, HPE Grad-CAM maps, and t-SNE embeddings of zero-shot matching features.

4.3 Ablation Study

We analyze the contribution of each PoseBridge component in Table 5. The baseline uses only skeleton sequences from the original pose estimator, without HPE intermediate features or pose-anchored semantics. On the ZSSAR side, the skeleton-conditioned semantic bridge (SB) improves performance by injecting temporal pose-anchored cues, while prototype adaptation (PA) further aligns text prototypes with the pose-semantic space. On the HPE side, hierarchical refinement (HR) and body-aware pooling (BP) provide complementary gains by preserving and selecting body-aware semantics. The full model performs best, confirming the benefit of combining HPE-side semantic preservation with ZSSAR-side semantic transfer.

Table 5: Ablation study of PoseBridge components on NTU-RGB+D 60 (48/12 split).

HPE		ZSSAR		NTU-RGB+D 60 (Xsub, 48/12 Split)	
HR	BP	SB	PA	ZSL	GZSL *(H)*
-	-	-	-	54.7	52.0
-	-	-	✓	56.6	57.3
-	-	✓	-	60.4	60.7
-	-	✓	✓	61.4	61.7
-	✓	✓	✓	64.0	63.2
✓	-	✓	✓	64.8	63.8
✓	✓	-	✓	66.9	63.9
✓	✓	✓	-	71.0	66.5
✓	✓	✓	✓	**73.2**	**68.0**

4.4 Qualitative Analysis

We qualitatively verify that PoseBridge recovers semantic cues that are largely missing from skeleton coordinates alone. In Fig. 4, we compare a baseline trained only with the original pose estimation objective and our HPE model trained with action-semantic alignment, hierarchical refinement, and body-aware pooling. For the confusing action pair "folding paper" and "typing on a keyboard", Grad-CAM shows that the baseline mainly responds to the overall human body, reflecting its bias toward keypoint localization. In contrast, our pose-anchored feature focuses more strongly on action-relevant regions, such as the keyboard, the paper, and hand-object interaction areas. The t-SNE plots show a consistent trend: baseline features are highly mixed, while PoseBridge yields much clearer separation between the two classes. These results show that PoseBridge captures action-relevant semantics during HPE and transfers them effectively to skeleton-text matching in zero-shot recognition.

5 Conclusion

We presented PoseBridge, an HPE-aware framework for ZSSAR that addresses semantic loss during skeletonization. Instead of treating human pose estimation as a fixed preprocessing step, PoseBridge preserves pose-anchored semantics from intermediate HPE representations and transfers them to skeleton-text alignment through a skeleton-conditioned semantic bridge and semantic prototype adaptation. Experiments on NTU-RGB+D 60/120, PKU-MMD, and Kinetics-200/400 show consistent improvements, especially under harder splits and in-the-wild settings. Importantly, PoseBridge achieves these gains without an additional RGB recognition branch, showing that the HPE process itself can serve as an effective upstream semantic source. Overall, our results suggest that stronger ZSSAR can be achieved by preserving and bridging the semantics already present within skeletonization, rather than adding heavier visual pathways after skeleton extraction.

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Appendix

This appendix provides additional details, results, and analyses that support the main paper. Section A describes the implementation details of the HPE and ZSSAR stages. Section B summarizes the datasets, seen/unseen split settings, and action descriptions. Section C reports additional performance comparisons, including challenging splits, Xview/Xset protocols, controlled comparisons with estimated 2D skeleton inputs, and the effect of semantic alignment on pose estimation. Section D analyzes key design choices, including hyperparameters, component-wise computational costs, and different pose estimators. Section E provides additional class-level analyses with confusion matrices. Section F presents additional qualitative results, including t-SNE and Grad-CAM visualizations. Section G discusses limitations and broader impacts.

A Implementation Details

A.1 HPE Details

We use RTMPose [11] as the 2D pose estimator and train it on MS COCO [17]. For the top-down pose estimation pipeline, we use YOLOv11 [13] as the human detector. Following the standard RTMPose-s setting, we use an input resolution of 256×192 and train the model for 420 epochs with AdamW. The base learning rate is set to 4×10^{-3}, the weight decay is set to 0, and the learning rate is adjusted with a linear warm-up followed by cosine annealing. The model is optimized with the original SimCC-based pose estimation loss and our action-semantic alignment loss described in Sec. 3.2.

For action-semantic alignment, we use COCO captions as image-level textual supervision. We encode each caption using a frozen CLIP ViT-B/32 text encoder [22] and map the resulting feature to the pose-semantic embedding space with a lightweight projection head. The projection dimension is set to 512, and the semantic loss weight is set to $\lambda_{\text{hpe}} = 0.1$. The CLIP-style contrastive loss for semantic alignment is applied from the beginning of training.

For the pose-anchored semantic cue extractor, we use $L = 3$ feature levels by default. The hierarchical refinement gate is fixed to $\alpha = 0.5$. After training, the HPE model is frozen and used to extract both 2D skeleton sequences and pose-anchored semantic cues for the ZSSAR stage. Additional analyses on design choices, including different pose estimators such as HRNet [27] and ViTPose [32], are reported in Appendix D.

A.2 ZSSAR Details

We use Shift-GCN [4] as the skeleton encoder and the pretrained CLIP ViT-L/14 text encoder [22] for semantic prototype extraction. The CLIP text encoder is frozen, and action descriptions are encoded and cached before training. The skeleton feature dimension is 256, the pose-semantic cue dimension is 512, and the shared embedding dimension is 512. The skeleton projector uses two MLP layers, while the pose adapter uses one MLP layer. The dropout ratio is set to 0.1.

The skeleton-conditioned semantic bridge uses multi-head cross-attention with 4 heads. For each video, we use 16 temporal pose-anchored cues. The skeleton representation is used as the query, and the temporal pose-anchored cues are used as keys and values. For prototype adaptation, we use the top-$K = 5$ nearest seen classes in the text-prototype space. The prototype adaptation coefficient is set to $\rho = 0.2$.

We train PoseBridge with AdamW for 30 epochs. The batch size is 128, the test batch size is 256, the learning rate is 1×10^{-3}, and the weight decay is 2×10^{-3}. We use 5 warm-up epochs followed by cosine annealing with a minimum learning rate of 1×10^{-6}. Gradient clipping is applied with a maximum norm of 1.0. We also use an exponential moving average (EMA) with decay 0.999 starting from epoch 5. All experiments are conducted on a single NVIDIA RTX A6000 GPU.

For the loss weights, we set $\lambda_s^{\text{cls}} = 1.0$, $\lambda_s^{\text{sem}} = 1.5$, $\lambda_s^{\text{con}} = 1.5$, $\lambda_p^{\text{cls}} = 0.5$, $\lambda_p^{\text{sem}} = 0.5$, $\lambda_p^{\text{con}} = 0.3$, $\lambda_{\text{s2p}} = 0.3$, $\lambda_{\text{kd}} = 1.0$, $\lambda_b^{\text{sem}} = 1.0$, and $\lambda_b^{\text{con}} = 0.5$. The supervised contrastive temperature is set to 0.07, and the distillation temperature is set to 4.0. Additional analyses on design choices are reported in Appendix D.

B Dataset Details

B.1 Dataset Details

NTU-RGB+D 60. NTU-RGB+D 60 [24] is a large-scale RGB-D action recognition benchmark containing 56,880 samples from 60 action classes performed by 40 subjects. The dataset was captured by three Kinect V2 cameras and provides multi-modal observations, including RGB videos, depth maps, infrared sequences, and 3D skeleton annotations. The standard evaluation protocols are cross-subject (Xsub) and cross-view (Xview), which evaluate generalization across different subjects and camera viewpoints, respectively. For zero-shot evaluation, we follow the seen/unseen class splits adopted in prior ZSSAR works [8, 34, 35].

NTU-RGB+D 120. NTU-RGB+D 120 [20] extends NTU-RGB+D 60 to 114,480 samples from 120 action classes. It is also captured with Kinect V2 sensors and provides RGB, depth, infrared, and skeleton modalities. Compared with NTU-RGB+D 60, it contains a larger action vocabulary and more diverse acquisition setups, making it a more challenging benchmark for zero-shot generalization. The standard evaluation protocols are cross-subject (Xsub) and cross-setup (Xset), where Xset evaluates generalization across different camera and environment setups. We follow the standard ZSSAR seen/unseen splits used in previous works [8, 34, 35].

PKU-MMD. PKU-MMD [18] is a multi-modal RGB-D benchmark for human action understanding captured with Kinect v2 sensors. Following prior ZSSAR protocols [8, 34, 35], we use the commonly adopted 51-class setting, which contains 1,076 long video sequences and nearly 20,000 action instances performed by 66 subjects from three camera views. The standard evaluation protocols are cross-subject (Xsub) and cross-view (Xview).

Kinetics-200/400. Kinetics-400 [12] is a large-scale in-the-wild video action recognition dataset collected from YouTube videos. Unlike the NTU-RGB-D 60/120 and PKU-MMD datasets above, Kinetics contains larger variations in scene, viewpoint, subject appearance, camera motion, and action context. Following the PURLS benchmark [35], Kinetics-200 is constructed from the first 200 classes of Kinetics-400 and evaluated under multiple seen/unseen splits. We additionally evaluate on Kinetics-400 using analogous splits to test scalability under a larger action vocabulary.

Skeleton extraction. For all datasets, we extract 2D skeletons from RGB frames using RTMPose [11]. We also obtain pose-anchored semantic cues from the same pose estimation process.

Auxiliary HPE training data. We train the 2D pose estimator on MS COCO [17]. In addition to the original pose estimation supervision, we use image-level textual descriptions from COCO captions as semantic supervision for the proposed action-semantic alignment loss. Specifically, the pose-anchored semantic cue extracted from the HPE representation is aligned with the corresponding caption feature encoded by the pretrained CLIP text encoder [22]. MS COCO is used only for training the HPE model and is not used for ZSSAR training.

B.2 Seen/Unseen Split Details

We follow the seen/unseen class splits used in prior ZSSAR protocols for fair comparison. For NTU-RGB+D 60, we evaluate the 55/5 and 48/12 splits; for NTU-RGB+D 120, we evaluate the 110/10 and 96/24 splits. For PKU-MMD, we use the standard 46/5 split adopted in previous ZSSAR works. For Kinetics-200, we follow the PURLS benchmark [35] and evaluate the 180/20, 160/40, 140/60, and 120/80 splits. For Kinetics-400, we use analogous 360/40, 320/80, 300/100, and 280/120 splits. All models are trained only on the seen classes of each split, and no samples from unseen classes are used during training.

B.3 Action Description Details

We use LLM-generated action descriptions to construct text prototypes for ZSSAR. Following the semantic setting introduced by SMIE [34], each action category label is expanded into a richer action description using ChatGPT. For example, the category label "drink water" is expanded into "the act of taking in water by mouth to hydrate or quench thirst as a human action." These action-level descriptions are encoded by the frozen CLIP text encoder [22] and used as semantic prototypes for both seen and unseen classes.

Table 6: Results under more challenging seen/unseen splits on NTU-RGB+D 60 and NTU-RGB+D 120. The best results are highlighted in **red**, and the second-best results are blue.

Method	Venue	NTU-RGB+D 60 (Xsub)		NTU-RGB+D 120 (Xsub)	
		40/20 Split	30/30 Split	80/40 Split	60/60 Split
ReViSE [9]	ICCV 2017	24.3	14.8	19.5	8.3
JPoSE [30]	ICCV 2019	20.1	12.4	13.7	7.7
CADA-VAE [23]	CVPR 2019	16.2	11.5	10.6	5.7
SynSE [8]	ICIP 2021	19.9	12.0	13.6	7.7
PURLS [35]	CVPR 2024	31.1	23.5	28.4	19.6
ScoPLe [36]	CVPR 2025	32.0	18.2	25.3	15.7
TDSM [5]	ICCV 2025	36.1	25.9	37.0	27.2
PoseBridge	**This work**	**41.4**	**37.2**	**43.7**	**30.4**

Table 7: Comparison on NTU-RGB+D 60 under the Xview protocol and NTU-RGB+D 120 under the Xset protocol (standard seen/unseen splits). We report ZSL accuracy and GZSL seen accuracy S, unseen accuracy U, and harmonic mean H.

Method	Venue	NTU-RGB+D 60 (Xview)								NTU-RGB+D 120 (Xset)							
		55/5 Split				48/12 Split				110/10 Split				96/24 Split			
		ZSL	GZSL			ZSL	GZSL			ZSL	GZSL			ZSL	GZSL		
		Acc	*S*	*U*	*H*	*Acc*	*S*	*U*	*H*	*Acc*	*S*	*U*	*H*	*Acc*	*S*	*U*	*H*
ReViSE [9]	ICCV 2017	54.4	25.8	29.3	27.4	17.2	34.2	16.4	22.1	30.2	4.0	23.7	6.8	13.5	2.6	3.4	2.9
JPoSE [30]	ICCV 2019	72.0	61.1	59.5	60.3	28.9	29.0	14.7	19.5	52.8	23.6	4.4	7.4	38.5	79.3	2.6	4.9
CADA-VAE [23]	CVPR 2019	75.1	65.7	56.1	60.5	32.9	49.7	25.9	34.0	52.5	46.0	44.5	45.2	38.7	47.6	26.8	34.3
SynSE [8]	ICIP 2021	68.0	65.5	45.6	53.8	29.9	61.3	24.6	35.1	59.3	58.9	49.2	53.6	41.4	46.8	31.8	37.9
SMIE [34]	ACMMM 2023	79.0	-	-	-	41.0	-	-	-	57.0	-	-	-	42.3	-	-	-
STAR [1]	ACMMM 2024	81.6	71.9	70.3	71.1	42.5	66.2	37.5	47.9	65.3	59.3	59.5	59.4	44.1	53.7	34.1	41.7
Neuron [2]	CVPR 2025	**87.8**	70.6	75.9	73.2	63.3	65.3	58.1	61.5	71.1	67.5	58.9	62.9	54.0	67.0	44.9	53.8
PoseBridge	**This work**	**87.8**	83.9	84.1	**84.0**	**65.9**	73.7	57.1	**64.4**	**77.1**	70.3	64.5	**67.3**	**65.0**	67.9	55.1	**60.8**

Recent ZSSAR methods further exploit LLM-generated semantics by constructing additional fine-grained descriptions, such as body-part-specific, temporal-phase-specific, or local/global action descriptions [35, 1, 31]. In contrast, PoseBridge does not introduce extra fine-grained LLM descriptions on the text side. We use the same action-level descriptions following SMIE and instead recover additional action-relevant semantics from the HPE process through pose-anchored semantic cues. This design ensures that the performance gain does not come from stronger text-side description engineering, but from the proposed pose-anchored semantic bridge.

C Additional Performance Comparisons

C.1 Results under More Challenging Seen/Unseen Splits

We further evaluate PoseBridge under more challenging seen/unseen splits, where the number of unseen classes is substantially increased. As summarized in Table 6, PoseBridge consistently achieves the best performance across all settings. The improvement is especially pronounced on the most challenging NTU-RGB+D 60 30/30 split, where PoseBridge improves the previous best result from 25.9% to 37.2%. These results indicate that recovering pose-anchored semantics is particularly useful when the model must distinguish among a larger and more diverse set of unseen action categories.

C.2 Results on Xview and Xset Protocols

We also test whether PoseBridge generalizes beyond the Xsub protocols used in the main paper. Table 7 reports additional results on NTU-RGB+D 60 Xview and NTU-RGB+D 120 Xset. PoseBridge achieves strong performance across both protocols and remains competitive or superior in both ZSL and GZSL settings. These results confirm that the proposed bridge is not limited to subject-based generalization, but also remains effective under viewpoint and setup shifts.

Table 8: Controlled comparison under the same estimated 2D skeleton input setting. All methods use the same RTMPose-extracted COCO-17 skeletons as input. We report ZSL accuracy and GZSL harmonic mean H on NTU-RGB+D 60/120 standard splits.

Method	Venue	NTU-RGB+D 60 (Xsub)				NTU-RGB+D 120 (Xsub)			
		55/5 Split		48/12 Split		110/10 Split		96/24 Split	
		ZSL	GZSL (*H*)	ZSL	GZSL (*H*)	ZSL	GZSL (*H*)	ZSL	GZSL (*H*)
Neuron [2]	CVPR 2025	88.3	61.1	58.8	49.0	66.5	48.2	60.2	45.6
TDSM [5]	ICCV 2025	86.9	-	64.7	-	76.3	-	59.7	-
FS-VAE [31]	ICCV 2025	87.8	32.7	48.8	41.8	62.6	46.9	56.2	48.5
PoseBridge	**This work**	**88.8**	**83.0**	**73.2**	**68.0**	**80.3**	**68.0**	**70.9**	**62.6**

Table 9: HPE-stage pose estimation quality and efficiency. We compare the standard RTMPose-s with the PoseBridge-enhanced RTMPose-s used to extract both skeletons and pose-anchored semantics. AP is evaluated on the MS COCO validation set, while parameters, GFLOPs, and FPS are measured for the HPE model.

Model	Training objective	AP	#Params (M)	GFLOPs (G)	FPS
RTMPose-s	$\mathcal{L}_{\text{pose}}$	70.86	5.47	0.68	4740.71
RTMPose-s + PoseBridge	$\mathcal{L}_{\text{pose}} + \mathcal{L}_{\text{sem}}^{\text{HPE}}$	70.24	8.23	0.83	2534.7

C.3 Fair Comparison with Estimated 2D Skeletons

Most prior ZSSAR methods report results using Kinect-based 3D skeletons with 25 body joints, whereas PoseBridge uses estimated 2D skeletons in the COCO-17 format extracted by RTMPose [11]. To ensure a fair comparison under the same skeleton input setting, we re-evaluate representative recent methods, including Neuron [2], TDSM [5], and FS-VAE [31], using the same RTMPose-extracted 2D skeletons as PoseBridge.

As shown in Table 8, PoseBridge remains consistently superior under this controlled setting. This confirms that the improvement does not come from an input-format advantage, but from the proposed pose-anchored semantic preservation and transfer mechanism. Even with the same estimated 2D skeleton inputs, PoseBridge effectively recovers action-relevant cues lost during skeleton extraction and transfers them to the semantic prototype space. This further demonstrates that our gains stem from improved skeleton-text alignment rather than stronger input representations.

C.4 Effect of Semantic Alignment on Pose Estimation

Since PoseBridge modifies the HPE model by adding semantic alignment during HPE training, we examine its effect on both pose estimation quality and HPE-stage efficiency. We compare the standard RTMPose-s setting with our HPE model trained using the action-semantic alignment objective in Sec. 3.2. We evaluate pose estimation quality using mean Average Precision (AP) on the MS COCO [17] validation set, and report the parameters, GFLOPs, and FPS of the pose estimator.

Different from Table 4, which measures only the ZSSAR recognition stage after skeleton extraction, Table 9 isolates the HPE cost. As shown in Table 9, our HPE model maintains comparable pose estimation quality to the original RTMPose-s [11], with AP changing only from 70.86 to 70.24. The added semantic alignment increases the HPE-stage cost moderately, from 0.68 to 0.83 GFLOPs, while preserving real-time throughput. This indicates that PoseBridge can learn pose-anchored semantic cues without substantially degrading pose estimation quality or HPE efficiency. Thus, the downstream ZSSAR gains are not caused by an improved skeleton format, but by more semantically useful HPE representations.

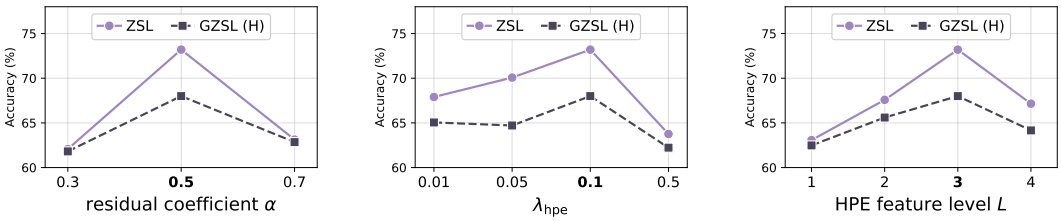


Figure 5: HPE-side hyperparameter analysis on NTU-RGB+D 60 under the standard 48/12 split. We vary the hierarchical refinement residual coefficient α, the semantic alignment weight λ_{hpe}, and the number of HPE feature levels L, while keeping the other settings fixed.

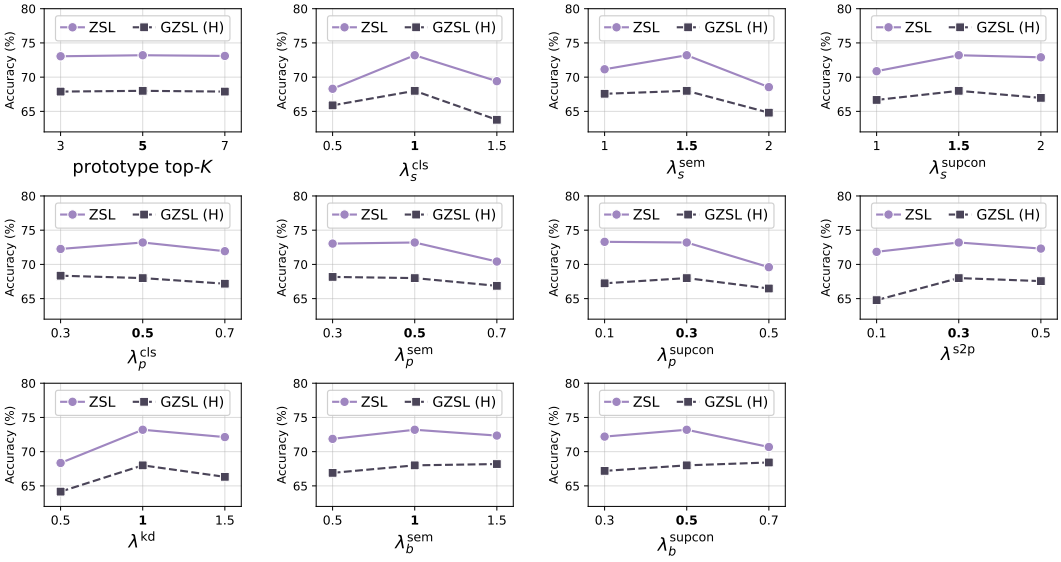


Figure 6: ZSSAR-side hyperparameter analysis on NTU-RGB+D 60 under the standard 48/12 split. We vary prototype top-K and key loss weights λ while keeping the other settings fixed.

D Additional Ablation and Design Studies

D.1 HPE-side Hyperparameters

We analyze the HPE-side hyperparameters used in Sec.3.2. Specifically, we vary the residual coefficient α in hierarchical refinement, the semantic alignment weight λ_{hpe}, and the number of HPE feature levels L. As shown in Fig. 5, the default setting $\alpha = 0.5$, $\lambda_{\text{hpe}} = 0.1$, and $L = 3$ provides the best overall performance. Using a small λ_{hpe} weakens semantic alignment, while an overly large value can disturb pose-oriented representation learning. Similarly, using too few HPE feature levels limits access to fine-grained visual cues, whereas adding more levels may introduce less action-relevant low-level responses. These results support our default HPE design for extracting pose-anchored semantic cues.

D.2 ZSSAR-side Hyperparameters

We further analyze the ZSSAR-side hyperparameters used in Sec.3.3 and Sec.3.4. Figure 6 shows the sensitivity to prototype adaptation, skeleton-side losses, pose-semantic losses, cross-branch regularization, and bridge-side losses. Overall, PoseBridge is stable across a reasonable range of hyperparameters, and the default values used in our experiments generally provide the best or near-best performance. The model is relatively insensitive to the prototype top-K, while loss weights related to semantic alignment and cross-branch transfer show clearer effects. This is consistent with

Table 10: Component-wise computational cost of the full PoseBridge online pipeline. The upper block reports shared pretrained encoders, while the lower block reports the cumulative cost from RTMPose-based skeleton and pose-anchored cue extraction to ZSSAR prediction. The baseline uses original RTMPose skeletons and performs skeleton-only ZSSAR without HPE intermediate features or pose-anchored semantics. GFLOPs and FPS are measured on a single NVIDIA RTX A6000 GPU.

Component	#Params (M)	GFLOPs (G)	FPS
Shared pretrained encoders			
Shift-GCN	0.8	0.778	–
CLIP Text Encoder	123.7	6.548	–
Cumulative online pipeline			
Baseline (with pretrained encoders)	131.0	8.009	1101.9
+HR	133.7	8.148	960.9
+HR+BP	133.7	8.153	940.8
+HR+BP+SB	136.4	8.168	935.3
+HR+BP+SB+PA (Full)	136.4	8.168	934.6

Table 11: Analysis with different HPE settings and 2D pose estimators. RTMPose (original) uses estimated skeletons without HPE intermediate features or pose-anchored semantic cues. For PoseBridge-HRNet, PoseBridge-ViTPose, and PoseBridge-RTMPose, we apply the same PoseBridge framework and report HPE performance, ZSSAR performance, parameters, GFLOPs, and FPS.

HPE Setting	MS COCO (AP)	NTU-RGB+D 60 (Xsub) 48/12 Split		NTU-RGB+D 120 (Xsub) 96/24 Split		#Params (M)	GFLOPs (G)	FPS
		ZSL	GZSL *(H)*	ZSL	GZSL *(H)*			
RTMPose (original)	70.9	54.7	52.0	55.1	52.4	**5.5**	**0.7**	**4740.7**
PoseBridge-HRNet [27]	70.8	65.3	63.8	**71.0**	61.7	28.9	7.7	766.9
PoseBridge-ViTPose [32]	**73.7**	69.2	61.8	69.8	61.7	766.9	6.2	1020.6
PoseBridge-RTMPose [11]	70.2	**73.2**	**68.0**	70.9	**62.6**	8.2	0.8	2534.7

our motivation: the final performance depends not only on skeleton-text matching, but also on how effectively pose-anchored semantics are transferred to the skeleton representation.

D.3 Component-wise Computational Cost

We report the component-wise computational cost of PoseBridge in Table 10. The table separates the shared pretrained encoders, i.e., Shift-GCN [4] and the CLIP ViT-L/14 text encoder [22], from the cumulative online pipeline, which starts from the skeleton-only baseline and incrementally adds each PoseBridge component. The baseline uses the original RTMPose for 2D skeleton extraction and performs skeleton-only ZSSAR without HPE intermediate features or pose-anchored semantics. The results show that hierarchical refinement (HR) and the skeleton-conditioned semantic bridge (SB) introduce the main additional parameters, while body-aware pooling (BP) and prototype adaptation (PA) add negligible overhead. Overall, PoseBridge improves recognition performance with moderate additional cost, supporting the efficiency of reusing HPE representations instead of adding a separate RGB recognition branch.

D.4 Different Pose Estimators

We evaluate whether PoseBridge is robust to the choice of the 2D pose estimator. We first include RTMPose-s (original) as a skeleton-only baseline, which uses estimated skeleton sequences without HPE intermediate features or pose-anchored semantic cues. In addition to our default RTMPose-s [11], we experiment with HRNet-w32 [27] and ViTPose-s [32]. Each pose estimator is trained on MS COCO using its standard pose-estimation recipe and hyperparameters, while applying the same HPE-side action-semantic alignment objective described in Sec.3.2. As shown in Table 11, PoseBridge consistently improves over the skeleton-only baseline and remains effective across different pose estimators. We also report MS COCO AP, parameter count, GFLOPs, and FPS for

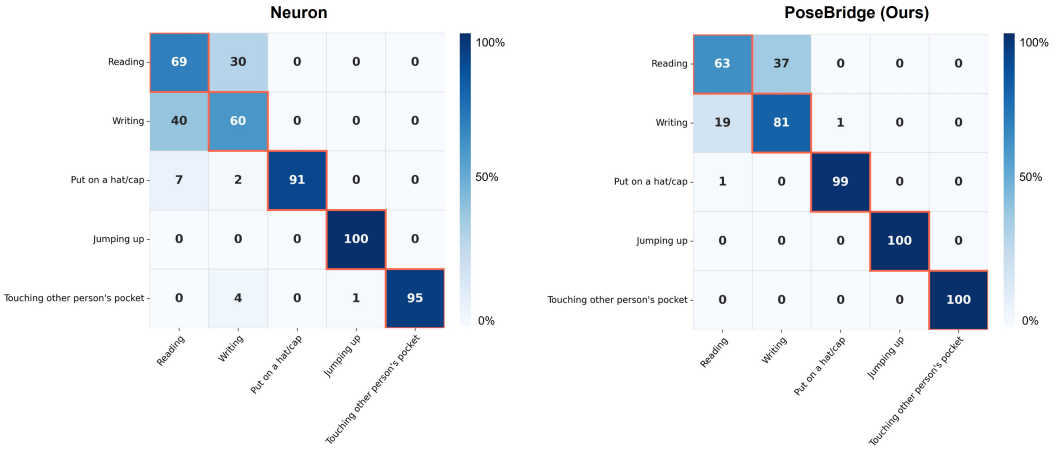


Figure 7: Row-normalized confusion matrices of Neuron and PoseBridge on NTU-RGB+D 60 under the standard 55/5 split.

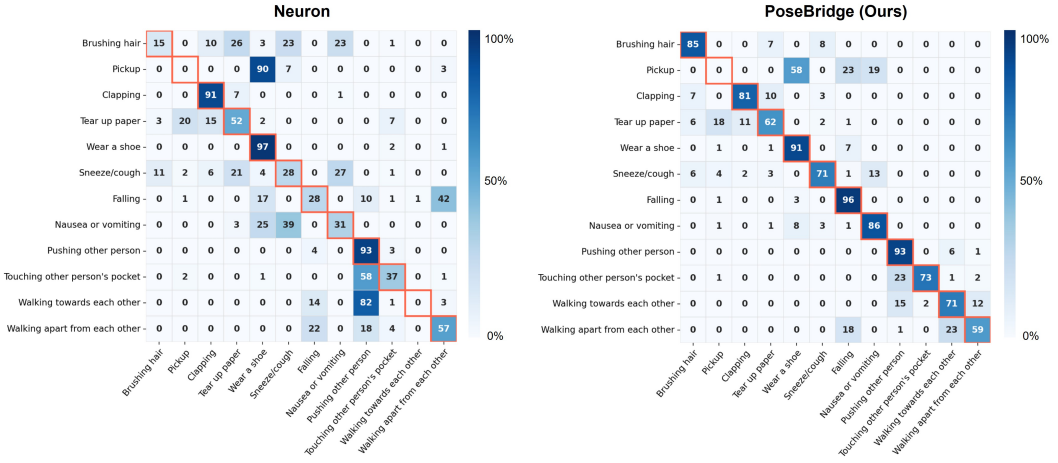


Figure 8: Row-normalized confusion matrices of Neuron and PoseBridge on NTU-RGB+D 60 under the standard 48/12 split.

reference. These results indicate that the proposed pose-anchored semantic bridge is not tied to a specific HPE backbone.

E Additional Analyses

We provide additional confusion matrix visualizations to analyze class-level prediction behavior under the same skeleton input setting. Specifically, we compare Neuron [2] and PoseBridge using RTMPose-extracted skeletons on the standard NTU-RGB+D 60/120 splits. As shown in Fig. 7–10, PoseBridge generally produces stronger diagonal responses and fewer off-diagonal confusions than Neuron. This trend is meaningful because both methods use the same estimated 2D skeleton inputs, indicating that the improvement comes from the proposed pose-anchored semantic bridge rather than differences in skeleton format. The results further show that pose-anchored semantics help distinguish semantically similar or motion-ambiguous actions in zero-shot recognition.

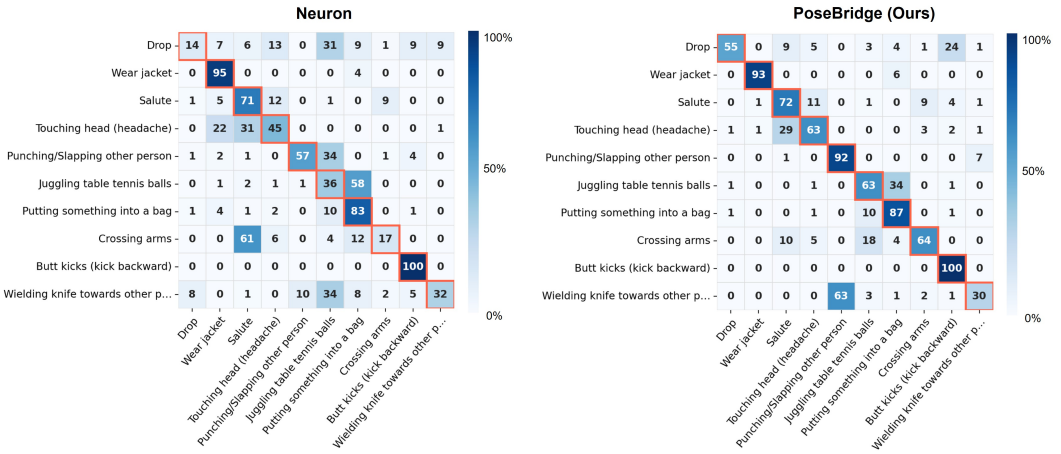


Figure 9: Row-normalized confusion matrices of Neuron and PoseBridge on NTU-RGB+D 120 under the standard 110/10 split.

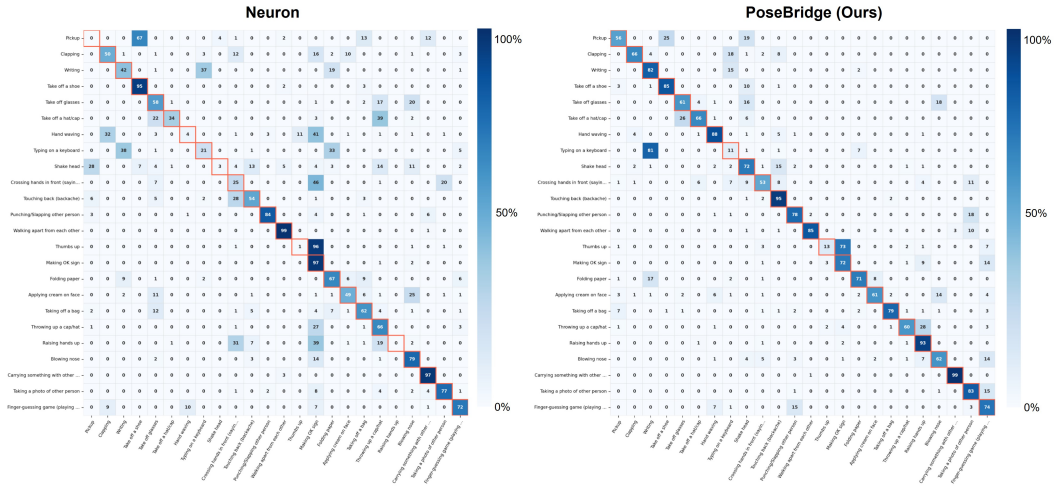


Figure 10: Row-normalized confusion matrices of Neuron and PoseBridge on NTU-RGB+D 120 under the standard 96/24 split.

F Additional Qualitative Results

F.1 t-SNE Visualization Against Neuron

To further analyze the representation quality of PoseBridge, we visualize zero-shot matching features using t-SNE and compare them with Neuron [2]. Both methods are evaluated under the same RTMPose-extracted 2D skeleton input setting, so the comparison focuses on the learned representation space rather than differences in skeleton format. As shown in Fig. 11 and Fig. 12, PoseBridge produces more compact intra-class clusters and clearer inter-class separation, whereas Neuron shows more scattered embeddings with noticeable overlap across classes. This indicates that the proposed pose-anchored semantics help organize skeleton features into a more discriminative semantic space. The trend is consistent on both NTU-RGB+D 60 under the 48/12 split and NTU-RGB+D 120 under the 96/24 split, suggesting that PoseBridge remains effective even when the unseen action space becomes larger and more ambiguous. These visualizations further support that bridging HPE-derived semantics to skeleton-text matching improves zero-shot class separability.

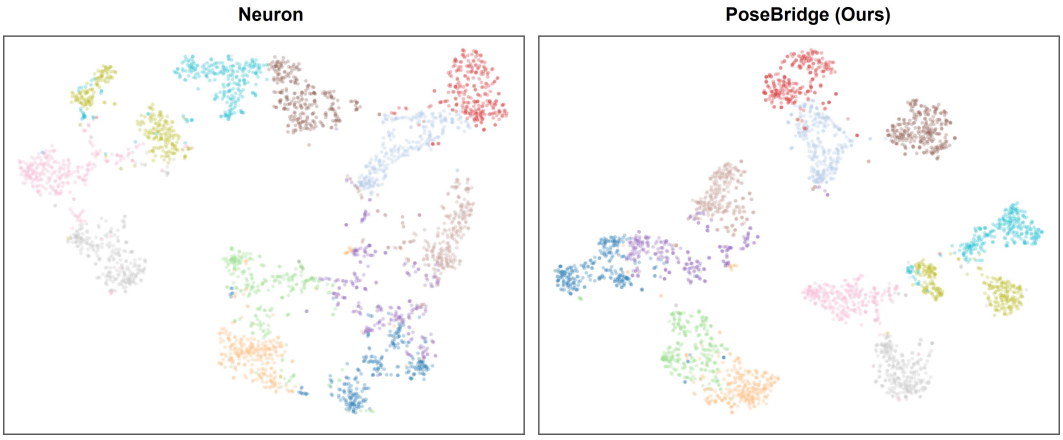


Figure 11: t-SNE visualization of zero-shot matching features on NTU-RGB+D 60 under the standard 48/12 split. We compare Neuron and PoseBridge using the same RTMPose-extracted 2D skeleton inputs.

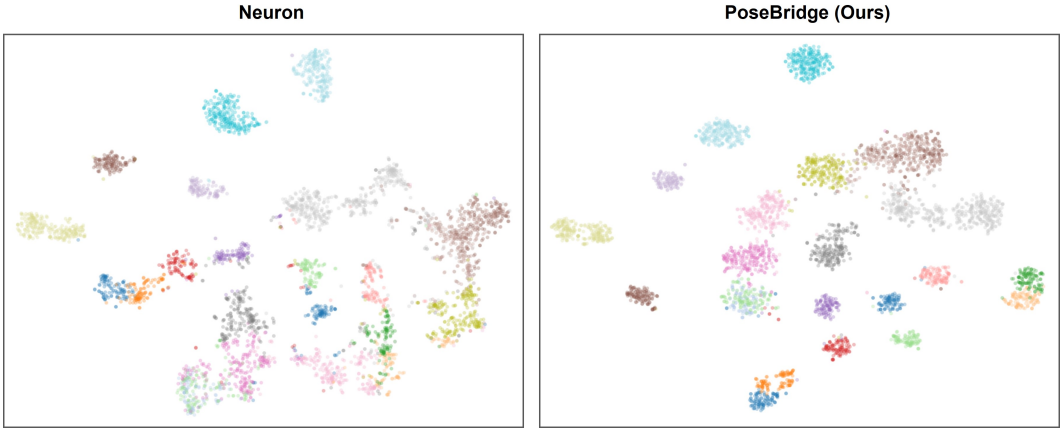


Figure 12: t-SNE visualization of zero-shot matching features on NTU-RGB+D 120 under the standard 96/24 split. We compare Neuron and PoseBridge using the same RTMPose-extracted 2D skeleton inputs.

F.2 Additional Grad-CAM Visualizations

We provide additional Grad-CAM examples in Fig. 13. The baseline HPE model, trained only for pose estimation, tends to activate broadly over the human body, reflecting its focus on keypoint localization rather than action semantics. In contrast, PoseBridge produces more localized and action-discriminative responses. For actions such as "pickup," "take off a shoe," "take off glasses," "take off a hat/cap," "play magic cube," "play with phone/tablet," and "writing," PoseBridge focuses more strongly on action-related visual cues around the manipulated object and the interacting body parts. For fine-grained gestures such as "make OK sign" and "make victory sign," it also highlights subtle hand and finger regions that are critical for distinguishing the action. These results further show that PoseBridge preserves action-relevant semantics during HPE, rather than relying only on coordinate-level skeleton information, which benefits downstream zero-shot skeleton-text matching.

G Limitations and Broader Impacts

Limitations. PoseBridge reduces the skeletonization gap by reusing pose-anchored semantic cues from the same HPE process that produces skeletons. Its effectiveness therefore depends on the quality

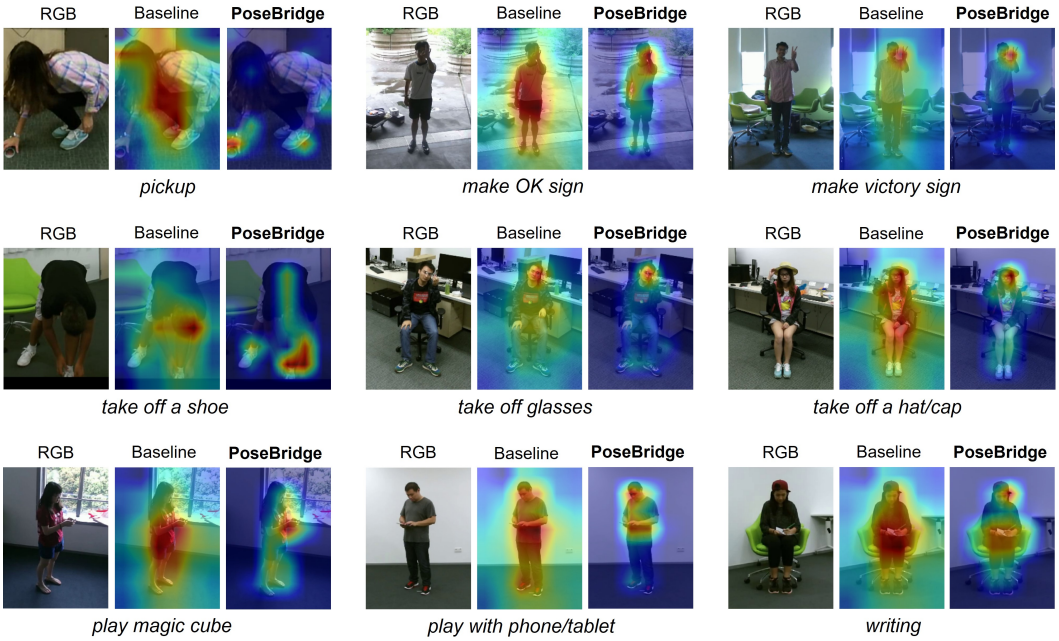


Figure 13: Additional qualitative comparison between the baseline and PoseBridge. We show RGB samples, Grad-CAM maps of baseline HPE features, and Grad-CAM maps of PoseBridge pose-anchored semantics for various actions.

and semantic coverage of the HPE stream. When the actor is heavily occluded, the manipulated object is very small or outside the body-attended region, or the pose estimator fails under severe motion blur or crowded scenes, both the skeletons and the pose-anchored cues may become unreliable. In addition, PoseBridge is designed for an HPE-aware ZSSAR setting rather than a strictly coordinate-only skeleton protocol, since it assumes access to intermediate HPE representations during skeleton extraction. Future work could further improve robustness under challenging visual conditions and investigate lighter ways to preserve pose-relative semantic cues.

Broader Impacts. PoseBridge aims to improve zero-shot skeleton action recognition, which can reduce the need for exhaustive action-specific annotation and may benefit applications such as human-computer interaction, rehabilitation analysis, sports motion understanding, and assistive robotics. However, action recognition systems can also raise privacy and fairness concerns when applied to real-world videos, especially in surveillance-like or high-stakes settings. Although PoseBridge does not introduce an independent RGB action-recognition branch, it still relies on videos during skeleton extraction and may inherit biases from the pose estimator, caption supervision, and pretrained language models. We therefore recommend careful dataset-specific validation, privacy-preserving deployment, and human oversight before using such systems in sensitive applications.